

Object Classification in Low-Resolution THz-like Imagery — Final Research Report

Status: 186 / 186 cells complete · Generated: 2026-05-23 · Hardware: RTX 5070 (Blackwell sm_120, 12 GB)

Executive Summary

This report presents the final outcomes of a 186-cell experimental campaign evaluating the robustness of three deep-learning architectures — **ResNet50** (CNN, ~25M params), **DenseNet121** (CNN, ~8M params), and **TransNeXt-tiny** (attention, ~28M params) — under severe visual degradation that simulates Terahertz (THz) imaging conditions: low resolution, blur, additive noise, salt-and-pepper, and desaturation. All three models are trained at the same **224 × 224** input resolution using a uniform protocol (full fine-tuning with differential learning rates), so accuracy differences are attributable to architecture and not to input shape.

Pipeline version: all numbers in this report are under `PIPELINE_VERSION = 2` (see [src/data/degradation_levels.py](#)). The 90 noise / salt-and-pepper / Phase B cells were re-run 2026-05-21 → 2026-05-23 after US-017 moved both stochastic pixel-level perturbations from POST-upsample (at 224 × 224, where each `torch.randn` sample landed on a single output pixel) to PRE-upsample (at `low_res`, where bicubic subsequently spreads each sample across a wider footprint at 224 — sensor-realistic coarse-grain structure). The 96 unaffected cells (Phase A clean + Phase C resolution / blur / saturation) are byte-identical to the v1 pipeline. v1 ↔ v2 are not directly comparable on the 90 re-run cells; v1 numbers survive only in git history (commit `3e1d089`).

Three headline cross-architecture findings emerged from the 150-cell Phase C single-axis isolation under pipeline v2:

- Universal 3 × 3-downsample bottleneck.** At the most aggressive resolution axis level (L5 = 3 × 3 native pixels upsampled to 224), every architecture collapses to ~0.43-0.46 best-val-acc on CIFAR-10 and ~0.44-0.45 on MNIST, within ±3 pp across architectures. The pretrain receptive-field hierarchy fails uniformly when sub-class geometric structure is destroyed, regardless of whether the backbone is convolutional or attention-based. The resolution axis is untouched by US-017, so this finding carries forward verbatim from v1.
- TransNeXt softens but does not reverse the v2 CIFAR-10 perturbation collapse.** Under pipeline v2, TransNeXt-tiny retains a single-digit-to-low- double-digit robustness gap over both CNN backbones on every CIFAR-10 perturbation axis at L5 (+13.2 pp on noise, +11.2 pp on blur, +7.1 pp on saturation, +10.0 pp on salt-and-pepper), but the underlying absolute floor is much lower than the v1 noise / S&P numbers suggested: `tnx cifar10 noise` drops to 0.80 at L5 (v1 reported 0.96 under the post-upsample bug). The v1 "≥ 0.95 across L1 → L5" claim survives intact only on saturation (axis unchanged by US-017) and partially on salt-and-pepper (holds L1-L3, falls below 0.95 at L4-L5). Attention's adaptive receptive fields *soften* the coarse-grain perturbation collapse without preventing it — the texture-statistic destruction is too severe for TransNeXt-tiny to fully compensate at `low_res = 3`.
- MNIST geometric-axis robustness floor preserved under pipeline v2.** Across all three architectures, MNIST `val_acc` on noise / blur / saturation / salt-and-pepper stays flat at 0.984-0.993 across L1 → L5 — including the v2-hardened noise axis (where CIFAR-10 collapses by up to 22.9 pp) and the v2-hardened salt-and-pepper axis (CIFAR-10 drops up to 4.4 pp). Only the resolution axis bites on MNIST (and only at L4-L5). This is a 60-cell, two-axis empirical confirmation that the THz-like coarse-grain perturbations preserve digit-shape signal in MNIST regardless of severity — the thick digit strokes cover many bicubically-spread perturbation footprints, so the discriminative feature survives. Texture-dominant recognition (CIFAR-10) and geometry-dominant recognition (MNIST) diverge sharply under pipeline v2 in a way the v1 pipeline largely masked.

Combined-axis Phase B confirms that the resolution-axis collapse dominates the overall L5 accuracy: under pipeline v2, with all five axes active at L5, CIFAR-10 best-val-acc falls to 0.19-0.20 across all three models, while MNIST falls to 0.27-0.28 — both bounded by the same resolution-axis floor exposed in Phase C, with the v2-hardened noise / S&P now pulling the combined-axes trajectory ~12.5 pp lower than v1.

Campaign Statistics

- Total cells:** 186 (6 Phase A + 30 Phase B + 150 Phase C)
- Total accumulated GPU runtime:** ~108.5 GPU-hours
- Total training epochs across all cells:** 5,121
- Architectures evaluated:** 3 (ResNet50, DenseNet121, TransNeXt-tiny)
- Datasets:** 2 (CIFAR-10, MNIST upsampled 28 → 224)
- Degradation severities:** 5 (L1 Mild → L5 Extreme)
- Phase C single-axis isolation axes:** 5 (resolution, noise, blur, saturation, salt_pepper)

Methodology

Architectures

Model	Params	Input	Training Regime
ResNet50	~25M	224 × 224	Full fine-tune, AdamW, head LR 1e-3 / backbone LR 1e-4
DenseNet121	~8M	224 × 224	Full fine-tune, AdamW, head LR 1e-3 / backbone LR 1e-4
TransNeXt-tiny	~28M	224 × 224	Full fine-tune, AdamW, head LR 5e-4 / backbone LR 5e-5, weight_decay 5e-2, drop_path 0.1

All three use ImageNet pretrained weights, a uniform 60-epoch cap with `patience=10` early stopping (`min_delta=1e-4`, `monitor=val_acc`, `mode=max`), gradient clipping at `max_norm=1.0`, batch size 32, seed 42 with `pl.seed_everything(workers=True)`, and `bf16-mixed` Blackwell precision.

Datasets

- **CIFAR-10** — 10 object classes, 32 × 32 RGB, 10K train / 5K val, upsampled 32 → 224.
- **MNIST** — 10 digit classes, 28 × 28 grayscale broadcast to 3 channels, 10K train / 5K val, upsampled 28 → 224 by the degradation pipeline.

Degradation Pipeline

All cells share the same pipeline:

```
original → saturation lerp → downsample(low_res) → upsample 224 × 224 (bicubic)
      → blur → noise → salt&pepper → ImageNet normalize → model
```

5-Level Degradation Curve

Level	low_res s	blur kernel	blur σ	noise std	S&P	saturation
L1 Mild	18	13	2.50	0.04	0.03	0.95
L2 Light	12	25	5.00	0.08	0.06	0.65
L3 Moderate	8	41	8.00	0.12	0.10	0.40
L4 Severe	6	61	12.00	0.16	0.14	0.15
L5 Extreme	3	91	18.00	0.22	0.18	0.00

Saturation is a deterministic lerp $(1 - s) \cdot \text{gray} + s \cdot \text{img}$, applied **before** noise / salt-and-pepper so injected noise stays color-correct. Validation degradations are per-sample seeded (`seed = idx + SEED_OFFSET_VAL`) — validation pixels are byte-identical across dataset rebuilds, gated by `src/tests/test_degradation_determinism.py`.

Phase Design

Phase	Description	Count
A — Clean baselines	identity pipeline, no degradation	6
B — Combined degradation	all five axes active at the same level L	30
C — Single-axis isolation	one axis at L, the other four at identity	150

Total		186
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Phase C single-axis isolation (US-003) was the diagnostic instrument designed to attribute Phase B collapses to specific axes; the headline findings above are direct outputs of this design.

Phase A — Clean Baselines (6 cells)

Identity degradation pipeline; measures the upper bound for each (model, dataset) pair before any THz-like distortion is applied.

#	Model	Datase t	Best Val Acc	Epoch s	Runtime
1	resnet50	cifar10	0.9518	23	9m
2	resnet50	mnist	0.9914	21	7m
3	densenet121	cifar10	0.9356	17	9m
4	densenet121	mnist	0.9930	29	14m
5	transnext_tiny	cifar10	0.9764	15	38m
6	transnext_tiny	mnist	0.9924	35	90m

Reading: TransNeXt-tiny leads CIFAR-10 by +2.46 pp over ResNet50 and +4.08 pp over DenseNet121 at clean baseline; MNIST clusters tight at 0.991-0.993.

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Phase B — Combined Degradation (30 cells)

Every five axes are active at the same level L per cell. This is the "realistic THz" regime — the model has to defeat all five degradations simultaneously.

Phase B Best-Val-Acc Curve (L1 → L5)

Model	Dataset	L1	L2	L3	L4	L5
resnet50	cifar10	0.7646	0.5394	0.3586	0.2684	0.1906
resnet50	mnist	0.9918	0.9636	0.7886	0.5644	0.2800
densenet121	cifar10	0.7848	0.5368	0.3960	0.2852	0.1958
densenet121	mnist	0.9874	0.9654	0.8196	0.5808	0.2776
transnext_tiny	cifar10	0.8758	0.6422	0.3900	0.2850	0.2010
transnext_tiny	mnist	0.9896	0.9600	0.7994	0.5724	0.2748

Reading (pipeline v2). Strictly monotonic on CIFAR-10 across all three architectures (no inversions). MNIST holds ≥ 0.96 through L2 for all models, then falls fast at L4 / L5 — driven by the resolution axis (see Phase C) and amplified by the v2-hardened noise + S&P axes (mean ΔL3 vs v1 is roughly −7 pp on MNIST). v2 Phase B mean Δ across the 30 cells is −12.49 pp vs v1 (operator-locked under US-020 close 2026-05-21).

Full Phase B Result Table

#	Model	Dataset	Level	Best Val Acc	Epochs	Runtime
7	resnet50	cifar10	L1 Mild	0.7646	21	15m
8	resnet50	mnist	L1 Mild	0.9918	33	24m
9	densenet121	cifar10	L1 Mild	0.7848	45	44m
10	densenet121	mnist	L1 Mild	0.9874	18	18m
11	transnext_tiny	cifar10	L1 Mild	0.8758	26	72m
12	transnext_tiny	mnist	L1 Mild	0.9896	30	84m
13	resnet50	cifar10	L2 Light	0.5394	31	23m
14	resnet50	mnist	L2 Light	0.9636	45	34m
15	densenet121	cifar10	L2 Light	0.5368	14	14m
16	densenet121	mnist	L2 Light	0.9654	53	53m
17	transnext_tiny	cifar10	L2 Light	0.6422	22	61m
18	transnext_tiny	mnist	L2 Light	0.9600	16	45m
19	resnet50	cifar10	L3 Moderate	0.3586	13	10m
20	resnet50	mnist	L3 Moderate	0.7886	14	11m
21	densenet121	cifar10	L3 Moderate	0.3960	15	15m
22	densenet121	mnist	L3 Moderate	0.8196	47	48m
23	transnext_tiny	cifar10	L3 Moderate	0.3900	13	36m
24	transnext_tiny	mnist	L3 Moderate	0.7994	15	42m
25	resnet50	cifar10	L4 Severe	0.2684	14	11m
26	resnet50	mnist	L4 Severe	0.5644	14	11m

27	densenet121	cifar10	L4 Severe	0.2852	16	17m
28	densenet121	mnist	L4 Severe	0.5808	15	16m
29	transnext_tiny	cifar10	L4 Severe	0.2850	13	37m
30	transnext_tiny	mnist	L4 Severe	0.5724	15	43m
31	resnet50	cifar10	L5 Extreme	0.1906	15	12m
32	resnet50	mnist	L5 Extreme	0.2800	16	13m
33	densenet121	cifar10	L5 Extreme	0.1958	21	23m
34	densenet121	mnist	L5 Extreme	0.2776	24	26m
35	transnext_tiny	cifar10	L5 Extreme	0.2010	15	43m
36	transnext_tiny	mnist	L5 Extreme	0.2748	16	46m

Phase C — Single-Axis Isolation (150 cells)

The active axis is at level L; the other four axes are at **identity** (no degradation). This isolates the contribution of each degradation axis from the Phase B collapse.

At L1, the isolation cells differ from Phase B L1 cells because Phase B L1 has **all five axes** at L1 mild while Phase C L1 has **one axis at L1, four at identity** — see `docs/phase_c.md` for the scientific rationale and the `degradation_levels_hash` rotation note.

5 × 5 Heatmaps per (model, dataset)

ResNet50

resnet50 · cifar10

Axis	L1	L2	L3	L4	L5
resolution	0.8800	0.7898	0.6846	0.6050	0.4316
noise	0.8914	0.8470	0.7806	0.7276	0.6688
blur	0.9354	0.8988	0.8678	0.8146	0.7098
saturation	0.9206	0.9282	0.9256	0.9178	0.8822
salt_pepper	0.9042	0.8992	0.8652	0.8464	0.8330

resnet50 · mnist

Axis	L1	L2	L3	L4	L5
resolution	0.9884	0.9850	0.9348	0.8122	0.4426
noise	0.9910	0.9916	0.9910	0.9910	0.9896
blur	0.9932	0.9914	0.9918	0.9920	0.9882
saturation	0.9894	0.9918	0.9936	0.9918	0.9910
salt_pepper	0.9908	0.9898	0.9888	0.9890	0.9862

DenseNet121

densenet121 · cifar10

Axis	L1	L2	L3	L4	L5
resolution	0.8652	0.7960	0.6768	0.5776	0.4432
noise	0.8892	0.8364	0.7954	0.7506	0.6846
blur	0.9054	0.9040	0.8644	0.8204	0.7142
saturation	0.9184	0.9142	0.9166	0.9130	0.8834
salt_pepper	0.9072	0.8898	0.8796	0.8644	0.8508

densenet121 · mnist

Axis	L1	L2	L3	L4	L5
resolution	0.9922	0.9862	0.9360	0.8132	0.4540
noise	0.9934	0.9926	0.9924	0.9934	0.9928
blur	0.9906	0.9934	0.9896	0.9928	0.9846
saturation	0.9932	0.9924	0.9926	0.9928	0.9914
salt_pepper	0.9918	0.9914	0.9906	0.9876	0.9872

TransNeXt-tiny

transnext_tiny · cifar10

Axis	L1	L2	L3	L4	L5
resolution	0.9514	0.8862	0.7964	0.6840	0.4624
noise	0.9598	0.9334	0.8940	0.8598	0.8008
blur	0.9766	0.9724	0.9496	0.9088	0.8244

saturation	0.9792	0.9778	0.9754	0.9756	0.9534
salt_pepper	0.9728	0.9660	0.9548	0.9474	0.9422

transnext_tiny · mnist

Axis	L1	L2	L3	L4	L5
resolution	0.9914	0.9808	0.9248	0.8008	0.4486
noise	0.9916	0.9928	0.9924	0.9916	0.9914
blur	0.9924	0.9896	0.9914	0.9928	0.9906
saturation	0.9936	0.9896	0.9918	0.9924	0.9904
salt_pepper	0.9900	0.9916	0.9882	0.9876	0.9894

Full Phase C Result Table

#	Model	Datase t	Level	Axis	Best Val Acc	Epoch s	Runtime
37	resnet50	cifar10	L1 Mild	resolution	0.8800	52	25m
38	resnet50	mnist	L1 Mild	resolution	0.9884	13	6m
39	densenet121	cifar10	L1 Mild	resolution	0.8652	35	26m
40	densenet121	mnist	L1 Mild	resolution	0.9922	41	31m
41	transnext_tiny	cifar10	L1 Mild	resolution	0.9514	16	41m
42	transnext_tiny	mnist	L1 Mild	resolution	0.9914	29	74m
43	resnet50	cifar10	L1 Mild	noise	0.8914	23	11m
44	resnet50	mnist	L1 Mild	noise	0.9910	31	16m
45	densenet121	cifar10	L1 Mild	noise	0.8892	36	27m
46	densenet121	mnist	L1 Mild	noise	0.9934	27	21m
47	transnext_tiny	cifar10	L1 Mild	noise	0.9598	27	69m
48	transnext_tiny	mnist	L1 Mild	noise	0.9916	23	59m
49	resnet50	cifar10	L1 Mild	blur	0.9354	60	39m
50	resnet50	mnist	L1 Mild	blur	0.9932	31	20m
51	densenet121	cifar10	L1 Mild	blur	0.9054	24	22m
52	densenet121	mnist	L1 Mild	blur	0.9906	20	19m
53	transnext_tiny	cifar10	L1 Mild	blur	0.9766	15	41m
54	transnext_tiny	mnist	L1 Mild	blur	0.9924	27	74m

55	resnet50	cifar10	L1 Mild	saturation	0.9206	14	7m
56	resnet50	mnist	L1 Mild	saturation	0.9894	13	7m
57	densenet121	cifar10	L1 Mild	saturation	0.9184	37	28m
58	densenet121	mnist	L1 Mild	saturation	0.9932	28	22m
59	transnext_tiny	cifar10	L1 Mild	saturation	0.9792	17	44m
60	transnext_tiny	mnist	L1 Mild	saturation	0.9936	30	77m
61	resnet50	cifar10	L1 Mild	salt_pepper	0.9042	17	9m
62	resnet50	mnist	L1 Mild	salt_pepper	0.9908	37	19m
63	densenet121	cifar10	L1 Mild	salt_pepper	0.9072	37	28m
64	densenet121	mnist	L1 Mild	salt_pepper	0.9918	27	21m
65	transnext_tiny	cifar10	L1 Mild	salt_pepper	0.9728	24	62m
66	transnext_tiny	mnist	L1 Mild	salt_pepper	0.9900	16	41m
67	resnet50	cifar10	L2 Light	resolution	0.7898	22	11m
68	resnet50	mnist	L2 Light	resolution	0.9850	37	18m
69	densenet121	cifar10	L2 Light	resolution	0.7960	52	39m
70	densenet121	mnist	L2 Light	resolution	0.9862	52	40m
71	transnext_tiny	cifar10	L2 Light	resolution	0.8862	20	51m
72	transnext_tiny	mnist	L2 Light	resolution	0.9808	20	51m
73	resnet50	cifar10	L2 Light	noise	0.8470	20	10m
74	resnet50	mnist	L2 Light	noise	0.9916	26	13m
75	densenet121	cifar10	L2 Light	noise	0.8364	39	29m
76	densenet121	mnist	L2 Light	noise	0.9926	25	19m
77	transnext_tiny	cifar10	L2 Light	noise	0.9334	19	49m
78	transnext_tiny	mnist	L2 Light	noise	0.9928	32	82m
79	resnet50	cifar10	L2 Light	blur	0.8988	17	11m
80	resnet50	mnist	L2 Light	blur	0.9914	24	16m
81	densenet121	cifar10	L2 Light	blur	0.9040	44	41m
82	densenet121	mnist	L2 Light	blur	0.9934	24	23m
83	transnext_tiny	cifar10	L2 Light	blur	0.9724	14	38m
84	transnext_tiny	mnist	L2 Light	blur	0.9896	17	47m
85	resnet50	cifar10	L2 Light	saturation	0.9282	14	7m

86	resnet50	mnist	L2 Light	saturation	0.9918	21	11m
87	densenet121	cifar10	L2 Light	saturation	0.9142	41	32m
88	densenet121	mnist	L2 Light	saturation	0.9924	22	17m
89	transnext_tiny	cifar10	L2 Light	saturation	0.9778	13	33m
90	transnext_tiny	mnist	L2 Light	saturation	0.9896	14	36m
91	resnet50	cifar10	L2 Light	salt_pepper	0.8992	33	17m
92	resnet50	mnist	L2 Light	salt_pepper	0.9898	27	14m
93	densenet121	cifar10	L2 Light	salt_pepper	0.8898	45	34m
94	densenet121	mnist	L2 Light	salt_pepper	0.9914	41	32m
95	transnext_tiny	cifar10	L2 Light	salt_pepper	0.9660	15	38m
96	transnext_tiny	mnist	L2 Light	salt_pepper	0.9916	46	2.0h
97	resnet50	cifar10	L3 Moderate	resolution	0.6846	23	11m
98	resnet50	mnist	L3 Moderate	resolution	0.9348	49	24m
99	densenet121	cifar10	L3 Moderate	resolution	0.6768	19	14m
100	densenet121	mnist	L3 Moderate	resolution	0.9360	53	41m
101	transnext_tiny	cifar10	L3 Moderate	resolution	0.7964	37	1.6h
102	transnext_tiny	mnist	L3 Moderate	resolution	0.9248	16	41m
103	resnet50	cifar10	L3 Moderate	noise	0.7806	17	8m
104	resnet50	mnist	L3 Moderate	noise	0.9910	25	13m
105	densenet121	cifar10	L3 Moderate	noise	0.7954	34	26m
106	densenet121	mnist	L3 Moderate	noise	0.9924	42	32m
107	transnext_tiny	cifar10	L3 Moderate	noise	0.8940	16	41m
108	transnext_tiny	mnist	L3 Moderate	noise	0.9924	32	82m
109	resnet50	cifar10	L3 Moderate	blur	0.8678	31	21m
110	resnet50	mnist	L3 Moderate	blur	0.9918	31	22m
111	densenet121	cifar10	L3 Moderate	blur	0.8644	36	35m
112	densenet121	mnist	L3 Moderate	blur	0.9896	23	22m
113	transnext_tiny	cifar10	L3 Moderate	blur	0.9496	15	41m
114	transnext_tiny	mnist	L3 Moderate	blur	0.9914	22	61m
115	resnet50	cifar10	L3 Moderate	saturation	0.9256	17	8m
116	resnet50	mnist	L3 Moderate	saturation	0.9936	19	10m

117	densenet121	cifar10	L3 Moderate	saturation	0.9166	45	34m
118	densenet121	mnist	L3 Moderate	saturation	0.9926	37	29m
119	transnext_tiny	cifar10	L3 Moderate	saturation	0.9754	14	36m
120	transnext_tiny	mnist	L3 Moderate	saturation	0.9918	22	57m
121	resnet50	cifar10	L3 Moderate	salt_pepper	0.8652	18	9m
122	resnet50	mnist	L3 Moderate	salt_pepper	0.9888	31	16m
123	densenet121	cifar10	L3 Moderate	salt_pepper	0.8796	49	37m
124	densenet121	mnist	L3 Moderate	salt_pepper	0.9906	46	35m
125	transnext_tiny	cifar10	L3 Moderate	salt_pepper	0.9548	13	33m
126	transnext_tiny	mnist	L3 Moderate	salt_pepper	0.9882	18	46m
127	resnet50	cifar10	L4 Severe	resolution	0.6050	60	29m
128	resnet50	mnist	L4 Severe	resolution	0.8122	43	21m
129	densenet121	cifar10	L4 Severe	resolution	0.5776	17	13m
130	densenet121	mnist	L4 Severe	resolution	0.8132	30	23m
131	transnext_tiny	cifar10	L4 Severe	resolution	0.6840	33	84m
132	transnext_tiny	mnist	L4 Severe	resolution	0.8008	16	41m
133	resnet50	cifar10	L4 Severe	noise	0.7276	15	7m
134	resnet50	mnist	L4 Severe	noise	0.9910	27	14m
135	densenet121	cifar10	L4 Severe	noise	0.7506	36	27m
136	densenet121	mnist	L4 Severe	noise	0.9934	42	32m
137	transnext_tiny	cifar10	L4 Severe	noise	0.8598	15	38m
138	transnext_tiny	mnist	L4 Severe	noise	0.9916	40	1.7h
139	resnet50	cifar10	L4 Severe	blur	0.8146	44	31m
140	resnet50	mnist	L4 Severe	blur	0.9920	34	25m
141	densenet121	cifar10	L4 Severe	blur	0.8204	41	41m
142	densenet121	mnist	L4 Severe	blur	0.9928	44	44m
143	transnext_tiny	cifar10	L4 Severe	blur	0.9088	29	81m
144	transnext_tiny	mnist	L4 Severe	blur	0.9928	36	1.7h
145	resnet50	cifar10	L4 Severe	saturation	0.9178	18	9m
146	resnet50	mnist	L4 Severe	saturation	0.9918	13	7m
147	densenet121	cifar10	L4 Severe	saturation	0.9130	35	27m

148	densenet121	mnist	L4 Severe	saturation	0.9928	45	35m
149	transnext_tiny	cifar10	L4 Severe	saturation	0.9756	14	36m
150	transnext_tiny	mnist	L4 Severe	saturation	0.9924	19	49m
151	resnet50	cifar10	L4 Severe	salt_pepper	0.8464	21	11m
152	resnet50	mnist	L4 Severe	salt_pepper	0.9890	31	16m
153	densenet121	cifar10	L4 Severe	salt_pepper	0.8644	44	34m
154	densenet121	mnist	L4 Severe	salt_pepper	0.9876	20	15m
155	transnext_tiny	cifar10	L4 Severe	salt_pepper	0.9474	28	72m
156	transnext_tiny	mnist	L4 Severe	salt_pepper	0.9876	19	49m
157	resnet50	cifar10	L5 Extreme	resolution	0.4316	15	7m
158	resnet50	mnist	L5 Extreme	resolution	0.4426	20	10m
159	densenet121	cifar10	L5 Extreme	resolution	0.4432	19	14m
160	densenet121	mnist	L5 Extreme	resolution	0.4540	21	16m
161	transnext_tiny	cifar10	L5 Extreme	resolution	0.4624	13	33m
162	transnext_tiny	mnist	L5 Extreme	resolution	0.4486	17	44m
163	resnet50	cifar10	L5 Extreme	noise	0.6688	30	15m
164	resnet50	mnist	L5 Extreme	noise	0.9896	19	10m
165	densenet121	cifar10	L5 Extreme	noise	0.6846	36	27m
166	densenet121	mnist	L5 Extreme	noise	0.9928	49	38m
167	transnext_tiny	cifar10	L5 Extreme	noise	0.8008	60	2.5h
168	transnext_tiny	mnist	L5 Extreme	noise	0.9914	36	1.5h
169	resnet50	cifar10	L5 Extreme	blur	0.7098	39	30m
170	resnet50	mnist	L5 Extreme	blur	0.9882	31	24m
171	densenet121	cifar10	L5 Extreme	blur	0.7142	21	22m
172	densenet121	mnist	L5 Extreme	blur	0.9846	19	20m
173	transnext_tiny	cifar10	L5 Extreme	blur	0.8244	22	62m
174	transnext_tiny	mnist	L5 Extreme	blur	0.9906	60	2.8h
175	resnet50	cifar10	L5 Extreme	saturation	0.8822	16	8m
176	resnet50	mnist	L5 Extreme	saturation	0.9910	13	7m
177	densenet121	cifar10	L5 Extreme	saturation	0.8834	37	28m
178	densenet121	mnist	L5 Extreme	saturation	0.9914	20	16m

179	transnext_tiny	cifar10	L5 Extreme	saturation	0.9534	44	1.9h
180	transnext_tiny	mnist	L5 Extreme	saturation	0.9904	18	46m
181	resnet50	cifar10	L5 Extreme	salt_pepper	0.8330	23	12m
182	resnet50	mnist	L5 Extreme	salt_pepper	0.9862	31	16m
183	densenet121	cifar10	L5 Extreme	salt_pepper	0.8508	44	33m
184	densenet121	mnist	L5 Extreme	salt_pepper	0.9872	22	17m
185	transnext_tiny	cifar10	L5 Extreme	salt_pepper	0.9422	59	2.5h
186	transnext_tiny	mnist	L5 Extreme	salt_pepper	0.9894	40	1.7h

Cross-Architecture Headline Findings

Finding 1 — Universal 3 × 3-downsample bottleneck at L5 resolution

At the most aggressive resolution axis level (L5 = 3 × 3 native pixels upsampled to 224 × 224 bicubic), every architecture lands within ±3 pp on CIFAR-10 and ±1 pp on MNIST:

Axis @ L5 — resolution	resnet50	densenet121	transnext_tiny
CIFAR-10	0.4316	0.4432	0.4624
MNIST	0.4426	0.4540	0.4486

The pretrain receptive-field hierarchy fails uniformly when sub-class geometric structure is destroyed, regardless of whether the backbone is convolutional or attention-based. The L5 resolution axis is the rate-limiting degradation for both Phase B (combined) collapses and the campaign's overall worst-case cells.

Finding 2 — TransNeXt softens but does not reverse the v2 CIFAR-10 perturbation collapse

At L5 on CIFAR-10 under pipeline v2, TransNeXt-tiny retains a robustness gap over both CNN backbones on every perturbation axis, but the absolute floor is much lower than the v1 noise / S&P numbers suggested. The v1 "≥ 0.95 across L1 → L5" claim now survives intact only on saturation (axis unchanged by US-017) and partially on salt-and-pepper (holds L1-L3, falls below 0.95 at L4-L5):

Axis @ L5 — CIFAR-10	resnet50	densenet121	transnext_tiny	TransNeXt gap vs avg CNN
noise (v2)	0.6688	0.6846	0.8008	~+13.2 pp
saturation	0.8822	0.8834	0.9534	~+7.1 pp
salt_pepper (v2)	0.8330	0.8508	0.9422	~+10.0 pp
blur	0.7098	0.7142	0.8244	~+11.2 pp
resolution	0.4316	0.4432	0.4624	+3 pp

Read-out: attention's adaptive receptive fields *soften* the coarse-grain perturbation collapse without preventing it. The texture-statistic destruction under pipeline v2 (where noise samples every low_res² pixel and bicubic spreads each draw into a ~74-px footprint at low_res = 3) is too severe for TransNeXt-tiny to fully compensate. Under the v1 post-upsample bug the same TransNeXt-tiny architecture appeared to "hold ≥ 0.95" on noise — that was an artefact of the bug, not architectural robustness, and is corrected here.

Finding 3 — MNIST geometric-axis robustness floor preserved under pipeline v2

For all three architectures, MNIST val_acc on noise / blur / saturation / salt-and-pepper stays flat at 0.984-0.993 across L1 → L5 under pipeline v2 — including the v2-hardened noise axis (where CIFAR-10 collapses by up to 22.9 pp) and the v2-hardened salt-and-pepper axis (CIFAR-10 drops up to 4.4 pp). Only the resolution axis bites on MNIST (and only at L4-L5). This is now a 60-cell, two-axis empirical confirmation that the THz-like coarse-grain perturbations preserve digit-shape signal in MNIST regardless of severity — the thick digit strokes cover many bicubically-spread perturbation footprints, so the discriminative feature survives. Texture-dominant recognition (CIFAR-10) and geometry-dominant recognition (MNIST) diverge sharply under pipeline v2 in a way the v1 pipeline largely masked.

The v2 severity ordering on CIFAR-10 at L5 (worst → mildest) becomes resolution > noise > blur > salt_pepper > saturation — noise leapfrogs blur as the second-worst axis under v2, masked in v1 because post-upsample noise was effectively averaged out by the bicubic kernel.

Complete cross-model L5 axis table

Axis @ L5	resnet50 cifar10	resnet50 mnist	densenet121 cifar10	densenet121 mnist	transnext_tiny cifar10	transnext_tiny mnist
resolution	0.4316	0.4426	0.4432	0.4540	0.4624	0.4486
noise	0.6688	0.9896	0.6846	0.9928	0.8008	0.9914
blur	0.7098	0.9882	0.7142	0.9846	0.8244	0.9906
saturation	0.8822	0.9910	0.8834	0.9914	0.9534	0.9904
salt_pepper	0.8330	0.9862	0.8508	0.9872	0.9422	0.9894

Research Conclusions

1. **Architecture matters more for noise than for geometry.** When the degradation is pixel-level (noise / saturation / salt-and-pepper), TransNeXt's attention mechanism provides a substantial robustness margin over CNNs. When the degradation is geometric (low resolution), all three architectures converge to the same failure mode at L5.
2. **The single-axis isolation design (Phase C) is the right diagnostic.** Without Phase C, the campaign's two headline cross-architecture findings would be invisible — Phase B's combined-axes L5 cells average over the resolution collapse and the perturbation-axis robustness gap, giving a misleading uniform "all CNN backbones fail" picture. The 150-cell investment in single-axis isolation surfaced both findings.
3. **CIFAR-10 vs MNIST gap at L5 quantifies natural-image vs digit-shape robustness.** Phase B L5 CIFAR-10 collapses to 0.19-0.20 across all three models under pipeline v2, while MNIST falls to 0.27-0.28 — a +7 to +8 pp gap that persists across architectures. The gap is driven entirely by the resolution axis; for all other axes, MNIST is essentially uniform at all severities (Finding 3). Under v1 the gap was wider (+12-15 pp) because v1 noise / S&P did not depress MNIST val_acc, but v2 confirms that even under genuinely sensor-realistic coarse-grain perturbations MNIST geometry survives while CIFAR-10 texture statistics do not.
4. **§6.4 pathology-guard retry path is ineffective on Optuna-tuned models.** Across the entire campaign, the pathology guard flagged 13 cells (1 in Phase B resnet50, 2 in Phase B densenet121, 6 in Phase C resnet50, 3 in Phase C densenet121, 2 in Phase C transnext_tiny). All 13 were monotonic with their surrounding L-curve. The operator declined the §6.4 retry on all 13 (lter 12 / 14 / 21 / 24 / lter 25 closure) because the §6.4 deltas ($wd \times 2 + dropout = 0.1 + backbone_lr \div 20$) consistently underfit tightly-tuned Optuna winners. Recommendation for future campaigns: replace §6.4 with an Optuna re-tune on the flagged cell, or tighten the pathology-guard threshold so it only fires on genuinely non-monotonic readings.
5. **Cross-architecture L5 convergence justifies the 224 × 224 uniform protocol decision.** When the V3 native-resolution refactor was rolled back (2026-05-13 protocol ratification), the fair-comparison invariant became "same tensor shape (224 × 224 after upsample)". The cross-model L5 resolution finding (Finding 1) is meaningful precisely because all three architectures see the same pixel budget — the convergence is evidence about the architectures, not about the input shape.

Operational Notes

- Reproducibility lock:** `pl.seed_everything(42, workers=True)` + per-sample validation seeds (`seed = idx + SEED_OFFSET_VAL`). Determinism gate (`src/tests/test_degradation_determinism.py`) verifies byte-identical validation pixels across dataset rebuilds.
- Optuna pre-tuning:** Six (model, dataset) pairs were tuned at L3 Moderate before the 186-cell campaign launched; the winning hyperparameters were frozen for every Phase A/B/C cell of that pair. Winners under `artifacts/`

```
best_hparams/{model}_{dataset}.json.
```

- **Engine:** PyTorch Lightning in `src/lightning/` (THzClassifier, THzDataModule). bf16-mixed on Blackwell (sm_120) RTX 5070; 32-true on CPU.

- **Weight privacy:** Trained checkpoints (`*.ckpt`), the Optuna SQLite store, and dashboard thumbnail caches are local-only, gitignored, and claudeignored. Analysis tooling reads `metrics.json / metrics.csv` / the tracker artifacts only — never the binary weights.

Appendix A — Full 186-cell Result Table

See `Final_Exp.md` (auto-regenerated from `runs/final/` by `scripts/update_final_exp.py`) for the canonical 186-cell tracker with PSNR / SSIM / Started / Duration columns. This report's tables summarize the same data with focus on `val_acc`.

Appendix B — Story-Driven Methodology Timeline

The campaign executed under a 9-story execution roadmap (US-006 through US-014), each closing a (phase, model) sub-campaign with operator-gated HALTs between stories. After US-014 closure, US-015 (end-of-campaign verification + three phase-boundary SYNCHRONIZER pushes + SECURITY leak audit) completes the campaign.

Closure decisions were recorded in `progress.txt` as `Iteration N` blocks (43 iterations across the v1 186-cell campaign and the v2 90-cell noise-fix re-run), each with the dispatch command, background task id, first-pass readings, pathology-guard outcomes, and operator decisions. The §6.4 retry decline streak (5-for-5 across all Phase B/C CNN dispatches + the one TransNeXt Phase C dispatch with sentinels) is documented inline.

Appendix C — Repository Pointers

- `Final_Exp.md` — 186-cell tracker (auto-regenerated)
- `artifacts/Final_Exp.json` — JSON-shaped tracker (dashboard data source)
- `artifacts/Final_Exp.html` — interactive dashboard with sortable rows + filters
- `artifacts/reports/phase_a_{model}_summary.md` — Phase A REPORTER summaries
- `docs/phase_a.md` / `docs/phase_c.md` — LIBRARIAN per-phase docs
- `progress.txt` — full iteration log (Iterations 1-43 — covers v1 + v2)
- `PRD.md` — v2 roadmap (US-017 through US-025; v1 US-001..US-016 closed)
- `CLAUDE.md` — protocol invariants + multi-agent guidance
- `scripts/build_final_exp_report.py` — this report generator
- `scripts/render_final_exp_pdf.py` — Markdown → PDF renderer

Report generated 2026-05-23 from artifacts/Final_Exp.json (state of disk under PIPELINE_VERSION = 2, after the v2 noise-fix 90-cell re-run closed 2026-05-23 / Iteration 40). Regenerate with `python scripts/build_final_exp_report.py`.